Lecture 1: Latent Representations

Empirical Research with Unstructured Data

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Oct 30, 2026













The world is unstructured.

- Economic
  - **Empirical research in social science studies high-dimensional concepts**: market power, the quality of products, institutions, policy uncertainty, the novelty of an invention, a judge's attitudes. None is observed directly, and no single proxy captures it.
  - **Data Revolution:** Digital traces (social media, satellite imagery, digitized archives) allow us to measure previously "invisible" economic behaviors with high granularity at scale.

- Goal: bridge between that advantage and modern machine learning methods for unstructured data
  - × Deploy machine learning systems as engineers who cares more about prediction(which becomes very feasible with current AI tools)
  - ✓ Understand the choice sets that modern AI technologies expand for economic research
    - ⋆ defining meaningful constructs
    - ⋆ designing credible empirical strategies
  - ♡ Let's be critical and effective users of these methods in economic research, given understanding data-generating processes

Outline of this Course

Oct 30	**Lecture 1** **Latent Representations**	• Text first: from the simplest bag-of-words representation to transformer-based LLMs • Trade-offs between textual representations • Applications to economic research
Nov 6	**Lecture 2** **LLM Agents**	• Autonomy and capabilities of LLM agents • LLMs as instruments for annotation and simulation, and their stability across prompts and models
Nov 13	**Lecture 3** **Econometric Problems, Interpretability, and (Imperfect) Solutions**	• Measurement error in ML-generated variables, with and without a validation set • Interpretability without ground truth: probes (is the information in the model?) and interventions (does the model rely on it?) • IRB issues
Dec 4	**Lecture 4** **Beyond Text: Other Modalities**	• Methods to other modalities of unstructured data

Course website: https://jiezhouecon.github.io/teaching/empirical-research-with-unstructured-data/

Outline

I. Measurement and Unstructured Data

II. Latent representations

III. Static word embeddings

IV. Contextual embeddings and the transformer

V. From representation to research

Revisit traditional pipelines for measurements

- Statistical methods operate on rectangular arrays, so empirical research has always relied on an *intermediate layer of infrastructure* that converts the raw record into tables:
  - Firm production $\rightarrow$ accounting ledgers $\rightarrow$ financial statements
  - Household spending $\rightarrow$ expenditure surveys $\rightarrow$ aggregate tables

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  - Firm production $\rightarrow$ accounting ledgers $\rightarrow$ financial statements
  - Household spending $\rightarrow$ expenditure surveys $\rightarrow$ aggregate tables
- Act of abstraction: choose which dimensions to record, design an encoding for them, then apply it (record, classify, aggregate)
  1. The dimensions are fixed *before* the data are collected, so information that was not anticipated is lost irretrievably.
  2. Each step has required specialized human labor (accountants, enumerators, statistical agencies), so the infrastructure is expensive to build and slow to change.

Measurement maps reality into a representation

Dell, "Measurement in the Age of AI," NBER Summer Institute Methods Lecture (2026)

- Measurement maps complex, high-dimensional reality into a lower-dimensional representation, preserving some features and discarding or transforming others. Formally:

$$X \xrightarrow{m(\cdot)} m(X)$$

  - A representation is useful only because it is simpler than what it represents
  - Selectivity does not make measurement merely subjective.

- Multiple ways: the object to have in mind is a family of candidate functions $m_j(X)$
- **Example: the population of a region.** A census, satellite imagery, mobile-phone records, and nighttime lights are four distinct m_j applied to the same underlying reality.
- To determine which m_j to use, the question is never "does this capture reality?" but:

Does this simplification preserve the features needed to support useful inference about the underlying high-dimensional world, **for the research question at hand**?

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2. Construct definition: *How should the concept be operationally defined?*
   - **Uncertainty:** how a phenomenon is translated into an operational construct for a given research setting
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3. Observation: *How can that definition be implemented at scale?*
   - **Uncertainty:** how accurately the construct can be observed at scale, and how errors propagate to the target parameter.
   - **Lenses:** measurement error; semiparametric inference with missing data.

Unstructured Data and Machine Learning Methods

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- Machine learning methods expand the choice set at each stage of $m(X)$ (Dell & Rambachan, 2026):
  1. **Discovery**: pre-specify a procedure rather than the construct. Embeddings, clustering, dictionary learning, and autoencoders surface dimensions nobody specified in advance, so we are not only looking under the lamppost.
  2. **Construct definition**: AI does not define the construct, but it makes candidate operationalizations (prompts, rubrics, dictionaries) cheap, along with the evidence for choosing among them.
  3. **Observation**: once the definition is fixed, AI applies it to millions of documents or images at low marginal cost, but its errors are systematic rather than classical.

The bottleneck shifts from building one scalable measure to choosing among many m_j, so validation becomes central.

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Representing Unstructured Data with High-Dimensional Formats

The Challenge: Machines do not process strings or sights; they process **tensors** (multi-dimensional arrays).

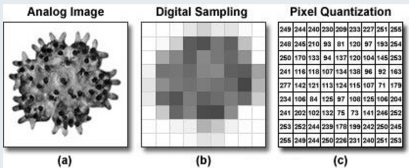
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Images: Spatial Tensors

Images are already numerical grids.

- **Structure:** $Height \times Width \times Channels$ (RGB).
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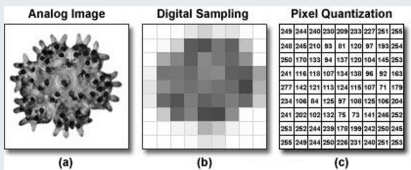
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Text: Sequential Tokens

Text must be discretized before mapping to a tensor.

- **Tokenization:** segment a sequence of text into discrete units called *tokens*.
- **Numerical Mapping:** Assigning a unique ID to each unit.

```
‘‘Trade policy matters’’
 → [1045, 3021, 2891]
```

Let’s start with text today.

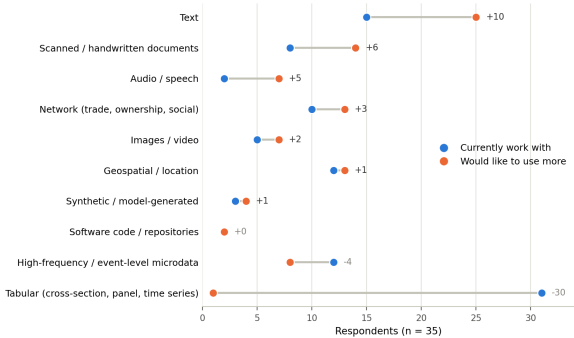
1. Text is where the data are.

   “An ever-increasing share of huma interaction, communication, and cu ture is recorded as digital text (Gentzkow, Kelly & Taddy, 2019).

   Class survey: strongest demand, widest gap.

2. From simple to modern.
   Counts → embeddings → transformers each built for language first, each to fix what the last could not. And you can rea text, so its limits show.

3. LLMs are models of text.



Pre-course survey: data types used now (blue) vs. wanted (orange).

Three families of latent representations

A **latent representation** turns labels into lower-dimensional vectors whose geometry means something. Three families, taken in order:

- High-Dimensional Count Vectors (Bag-of-Words): Representing documents as sparse vectors over a fixed vocabulary.
- Distributed Word Representations (Static Embeddings): Mapping tokens to dense vectors that capture semantic "neighborhoods."
  - *Word2Vec:* Mikolov et al. (2013); *GloVe:* Pennington, Socher, and Manning (2014)
- Contextualized Sequence Embedding (Dynamic Embeddings): Encoding meaning based on surrounding syntax (e.g., Transformers).
  - *ELMo:* Peters et al. (2018); *BERT:* Devlin et al. (2019); *GPT:* Radford et al. (2018)

- Topic modeling [Unsupervised learning]:
  - *Latent Dirichlet Allocation (LDA):* Blei, Ng, and Jordan (2003)

The Bag-of-Words (BoW) Model

The Concept: Documents are sets of word counts; syntax and order are discarded.

- **Mathematical Form:** A document d is a **sparse vector** $\mathbf{V}_d$.
- **The Basis:** Every word in the vocabulary V is a dimension in space.

The Atomic Unit: One-Hot Encoding

Each word w is a vector of length $|V|$ with a single 1 at its unique index:

$$\mathbf{v}_{\text{Tariff}} = [1, 0, 0, \ldots, 0], \quad \mathbf{v}_{\text{Trade}} = [0, 1, 0, \ldots, 0]$$

Implication: $\mathbf{v}_{\text{Tariff}} \cdot \mathbf{v}_{\text{Trade}} = 0$ (No semantic relationship).

A document vector: the sum of its one-hot word vectors: $\mathbf{V}_d = \sum_{w \in d} \text{count}(w) \cdot \mathbf{v}_w$

Document	*Tariff*	*Trade*	*Subsidy*	*Export*
Policy A	5	10	0	2
Policy B	0	8	12	1

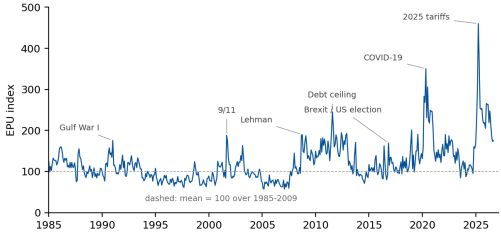
- Back to the three stages of $m(X)$:
  1. *discovery*: fixing the vocabulary
  2. *construct definition*: one-hot encoding
  3. *observation*: the document–term matrix

- Nothing is hidden inside a learned model, so every limitation is in plain view.
  1. **Dimensionality.** $|V|$ runs to millions and each document touches a vanishing fraction, so the vectors are sparse and cheap.
  2. **Order.** "The tariff did *not* raise prices" and "the tariff did raise prices" have nearly identical bags. Whether this matters is a question about the construct: fine for *how much is this document about trade policy* (the Baker–Bloom–Davis uncertainty index); fatal for *did the central bank signal tightening*.
  3. **Meaning.** One-hot vectors are orthogonal: *tariff* sits as far from *trade* as from *zucchini*.

A bag that works: the Economic Policy Uncertainty index

Baker, Bloom & Davis (2016)

- **Dictionary, built by hand:** an article in one of 10 US newspapers counts if it contains *uncertain(ty)* $\wedge$ *economic/economy* $\wedge$ one of {*Congress*, *deficit*, *Federal Reserve*, *legislation*, *regulation*, *White House*}.
- **Index:** monthly share of matching articles per paper, standardized, averaged across papers; mean 100 over 1985–2009.
- **Validation:** a 12,000-article human audit agrees with the count; spikes at known events; policy-exposed firms show higher stock volatility and lower investment and hiring.



Monthly US index. Source: FRED, USEPUINDXM.

Why the bag's blind spots do not bite

The construct is coarse (*is this article about policy uncertainty?*), so negation and order rarely flip the answer. The cost is the fixed vocabulary.

Beyond Counting: TF-IDF

The Goal: Give more weight to words that are “informative” and less to words that are common across all documents.

TF-IDF Weighting

The weight of a term v in document d is:

$$\text{TF-IDF}_{d,v} = \underbrace{x_{d,v}}_{\text{TF}} \times \underbrace{\log\left(\frac{D}{\sum_{j=1}^{D} \mathbb{I}(x_{j,v} > 0)}\right)}_{\text{IDF}}$$

$x_{d,v}$: raw count of term v in document d; D: number of documents; $\mathbb{I}(x_{j,v} > 0)$: indicator that term v appears in document j.

- **Intuition:** if “Tariff” appears often in one document but rarely in the corpus, it gets a **high** score. Stop words and ubiquitous domain words are filtered out automatically.
- **But:** TF-IDF changes the *weights* on the axes, not the axes. Each word is still its own orthogonal dimension.

Measuring Technological Innovation

“Measuring technological innovation over the long run.” (Kelly et al., 2021)

- **The Challenge:** Existing measures like patent citations have significant gaps:
  - Citations: not consistently recorded by the USPTO until 1947; rely on the discretion of inventors or examiners.

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- **The Challenge:** Existing measures like patent citations have significant gaps:
  - Citations: not consistently recorded by the USPTO until 1947; rely on the discretion of inventors or examiners.
  - Stock market data: only readily available post-1927
- **The Solution:** Use textual analysis of 9 million+ patent documents to create a text-based indicator of innovation.
- Issue with standard TF-IDF: “Forward-Looking” and penalizes breakthroughs.
  - **Example: Tesla (1888)**

    Term: “Alternating current.”

    Standard IDF: Sees this term in thousands of 20th-century patents $\rightarrow$ low weight.

    Result: A revolutionary invention appears “commonplace”.

Step 1: Retrospective Vectorization (TF-BIDF)

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Backward-IDF (BIDF)

$$BIDF_{wp} = \log\left(\frac{\#\text{ patents prior to } p}{1 + \#\text{ documents prior to } p \text{ containing term } w}\right)$$

Intuition: Word importance is measured only against the history leading up to the patent, not beyond it.

- Similarity between the patent pair (i, j)
  - TF-BIDF for each term w in patent i as

$$TFBIDF_{w,i,t} = TF_{w,i} \times BIDF_{w,t} \Rightarrow V_{i,t} = \underbrace{[TFIDF_{1,i,t}, ..., TFIDF_{W,i,t}]}_{1 \times W}$$

  - The cosine similarity between the two normalized vectors:

$$\rho_{i,j} = V_{i,t} \cdot V_{j,t}$$

Step 2: Defining the Quality Metric

- **Novelty $\rightarrow$ Backward Similarity (BS):**
  - Cosine similarity to patents in the **prior 5 years**.
  - Low BS = Conceptually distinct from predecessors.

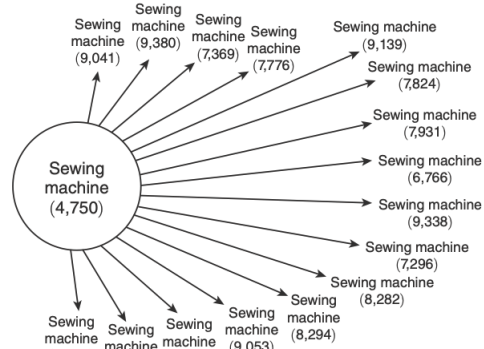
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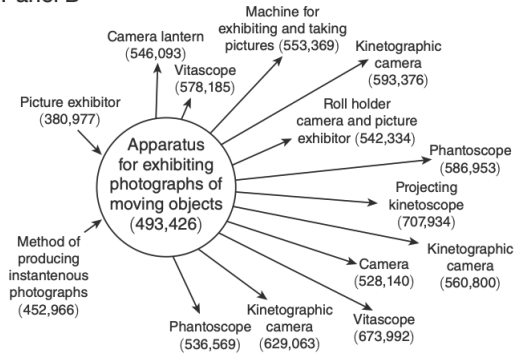
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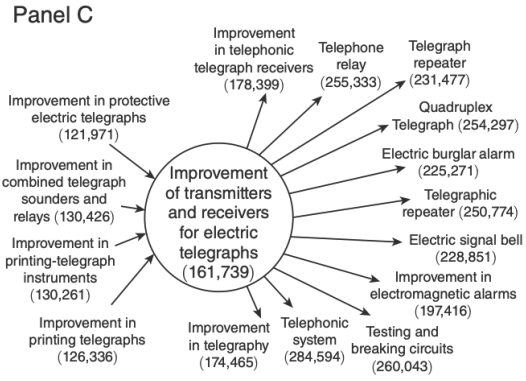


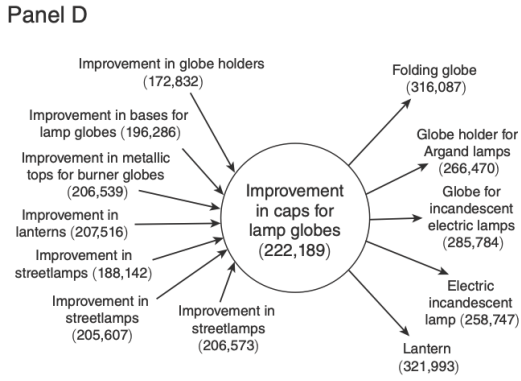
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Quality Indicator (q_j)

An important patent is both **novel** and **impactful**:

$$q_j = \frac{FS_j}{BS_j}$$

Breakthroughs: Top 10% of the q_j distribution $\rightarrow$ # of breakthrough inventions granted in each year, divided by US population

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- **Comparison Windows:** May underweigh "slow-burn" impact due to computational limits.
- **Sectoral Shifts:** Rising patenting rates (e.g., software) can inflate impact scores.
- **Data Scope:** Excludes non-patent literature (e.g., scientific journals).

Validation: Textual Innovation Measure vs. Benchmarks

- **Citations and value**: more important patents receive more forward citations and have higher stock-market value (KPSS, Kogan et al., 2017).

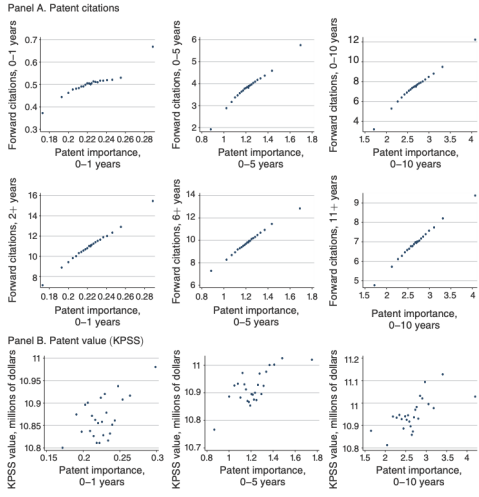


FIGURE 3. PATENT IMPORTANCE: VALIDATION

Figure: Forward citations (A) and KPSS value (B) vs. importance. Kelly et al. (2021), Fig. 3

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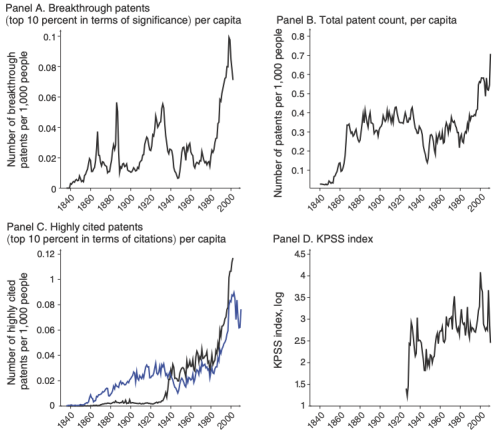


FIGURE 4. TECHNOLOGICAL INNOVATION OVER THE LONG RUN: NEW VERSUS EXISTING INDICATORS

Figure: Breakthrough patents per capita vs. existing indicators. Fig. 4

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- **Historical waves**: the aggregate index picks out the known waves (1870s–80s, 1920s–30s, post-1980).
- **By industry**: breakthroughs shift from manufacturing and transport (19th c.) to electrical and chemicals (mid-20th) to computers and electronics (late 20th).

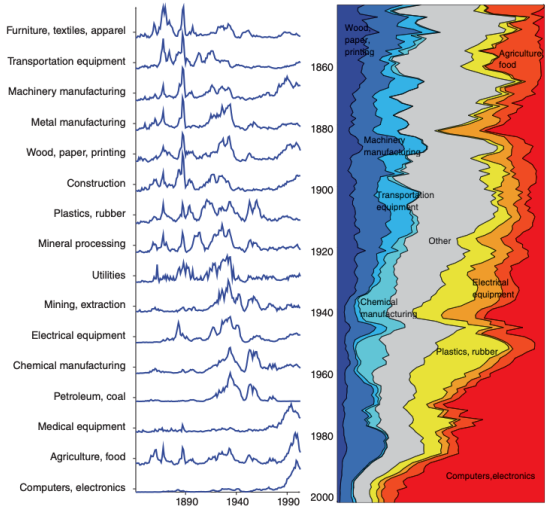


FIGURE 5. BREAKTHROUGH INNOVATION ACROSS INDUSTRIES

Figure: Breakthrough innovation by industry. Fig. 5

Discussion

1. Discovery
Which features matter?

The feature: a patent's vocabulary relative to its contemporaries. A breakthrough uses terms predecessors lacked and successors adopt.

Discarded: word order, synonyms, drawings, the scientific literature.

2. Construct definition
What is "importance"?

Theory: important = *novel* (unlike predecessors) and *impactful* (like successors).

Operational: ρ_{ij} = cosine of TF-BIDF vectors; BS over the prior 5 years, FS the next 10; $q = FS/BS$; breakthrough = top 10%.

The economics: backward IDF, so weights use only what was known at the time. Windows and cutoff: judgment calls.

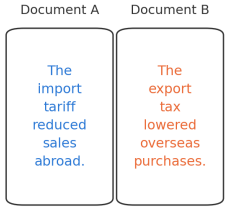
3. Observation
Does it hold at scale?

Implementation: vectorize every patent; similarities in rolling windows; count breakthroughs per year per capita and by industry.

Error sources: OCR of old text; growing sectors inflate FS; slow-burn inventions fall outside the window.

The measure works because the weighting scheme is designed around the economic concept: novelty is defined relative to *what was known at the time*, and that requires

Limitation of BoW: one orthogonal axis per word



Word	TF–IDF weight	
	Doc A	Doc B
the	0	0
import	0.050	0
tariff	0.050	0
reduced	0.050	0
sales	0.050	0
abroad	0.050	0
export	0	0.050
tax	0	0.050
lowered	0	0.050
overseas	0	0.050
purchases	0	0.050

Cosine similarity (Doc A, Doc B) = 0

One-hot word vectors

tariff	*tax*
0	0
0	0
1	0
0	0
0	0
0	0
0	0
0	**1**
0	0
0	0
0	0

tariff · tax = 0

Discrete symbols cannot see that an import tariff and an export tax are the same policy (Lerner, 1936).

- The two documents make the same claim about real outcomes:
  By Lerner symmetry (Lerner, 1936), a uniform import tariff and a uniform export tax at the same rate have identical real effects: each raises the domestic relative price of importables and, with balanced trade, reduces both imports and exports.

→ One fix: a representation in which **words used in the same contexts sit near each other**.

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Distributed Representations: Learning by Association

- **The Distributional Hypothesis:**

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The Training Objective (Skip-gram, Word2Vec)

Each word's meaning is a vector of parameters, chosen to predict the words within m positions of each target w_t:

$$J(\theta) = \frac{1}{T}\sum_{t=1}^{T} \sum_{-m \le j \le m,\, j \neq 0} \log P(w_{t+j} \mid w_t), \qquad P(o \mid c) = \frac{\exp\left(\mathbf{u}_o^\top \mathbf{v}_c\right)}{\sum_{w \in V} \exp(\mathbf{u}_w^\top \mathbf{v}_c)}$$

$\mathbf{v}_c$: vector of the center word; $\mathbf{u}_o$: vector of a context word.

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$\mathbf{v}_c$: vector of the center word; $\mathbf{u}_o$: vector of a context word.

- **Intuition:** $P(o \mid c)$ is high when the two vectors align, so training pulls "Tariff" and "Trade" closer; words sharing contexts end up nearby even if they never co-occur.
  Tariff and *quota* are rarely in the same sentence, but appear around the same words: *imports*, *protection*, *retaliation*.

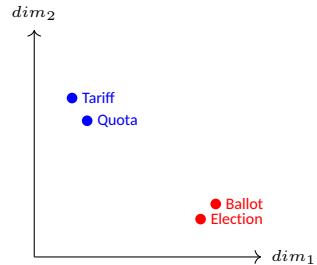
Distributed Representations: Word Embeddings

- **Dense Vectors:** Each word is represented by a vector of real numbers (e.g., length 300) rather than a sparse one-hot vector.

Geometry of Meaning

- **Synonyms:** Words with similar contexts are mapped to nearby points in the vector space.
- **Analogies:** Vector math works:

$$\vec{Beijing} - \vec{China} + \vec{USA} \approx \vec{Washington}$$

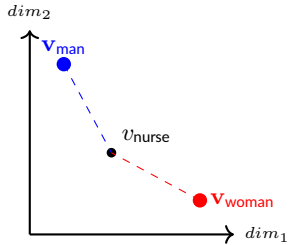


- **Relations are directions:** if *male vs. female* is a direction, we can measure how any word (an occupation, an adjective) leans along it.
- Innovation, construct exposure (product similarity, technology and labor (Kogan, Papanikolaou, Schmidt, Seegmiller, "Technology and Labor Displacement"),),

Quantifying Stereotypes: “Women Bias”

“Word embeddings quantify 100 years of gender and ethnic stereotypes.” (Garg et al., 2018)

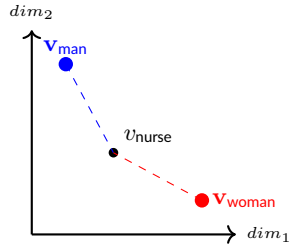
1. Defining the **Word Lists**
   - **Group Words (G):** gendered terms (*she*, *female* vs. *he*, *male*), averaged into $\mathbf{v}_{\text{woman}}$, $\mathbf{v}_{\text{man}}$
   - **Neutral Words (M):** terms to test for bias (*nurse*, *engineer*)



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2. **Metric**: Relative Norm Difference

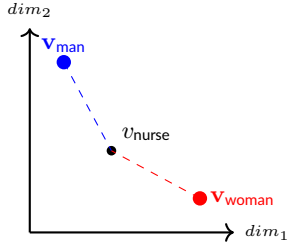
$$\text{Bias}(M) = \sum_{v_m \in M} (\underbrace{||v_m - \mathbf{v}_{\text{man}}||_2}_{\text{Dist. to Men}} - \underbrace{||v_m - \mathbf{v}_{\text{woman}}||_2}_{\text{Dist. to Women}})$$

   - **Positive:** Women-associated.
   - **Negative:** Men-associated.

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   - **Positive:** Women-associated.
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3. **Validation:** occupation scores line up with Census female shares ($r^2 = 0.499$); decade by decade, a **time series of attitudes**.

Measure the Gender Attitude at the Individual Level

"Gender Attitudes in the Judiciary: Evidence from U.S. Circuit Courts"(Ash et al., 2022)

1. **Train model (GloVe) from strach for each judge:**
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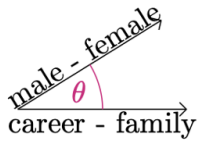
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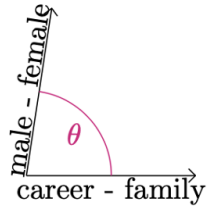
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$$\text{Slant}_j = \cos(\theta) = \frac{\vec{G} \cdot \vec{S}}{\|\vec{G}\|\|\vec{S}\|}$$

(a) Slant ≈ 1



(b) Slant ≈ 0



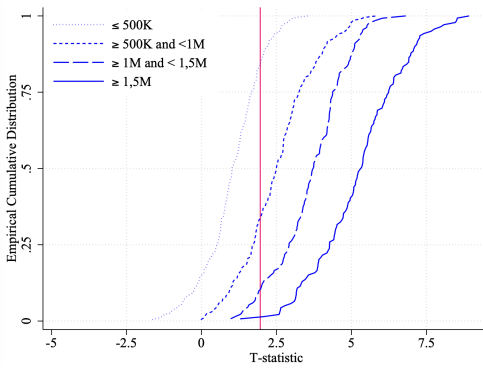
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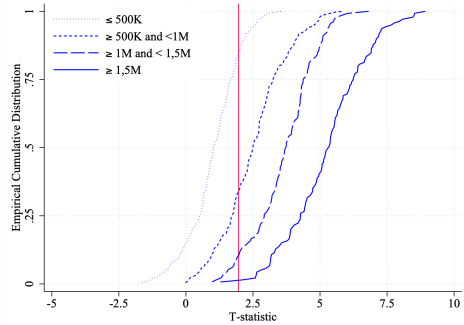
4. **Validation**: each judge-specific model correctly "learned" gender by testing name associations at the individual level



Validation, and from measure to behavior

Ash, Chen & Ornaghi (2024), continued

- **Does each judge-specific model actually learn gender?** Take common male and female first names from the Census, compute each name's similarity to the judge's gender dimension, and regress a male-name indicator on it, judge by judge.
- The test passes for most judges, and more reliably the more text a judge has written.
- **Use:** judges with higher slant vote more conservatively in gender-related cases and treat female colleagues differently.
- **The pipeline:** a latent attitude, measured from text through geometry, validated against an external benchmark, then used as a variable in an economic analysis.

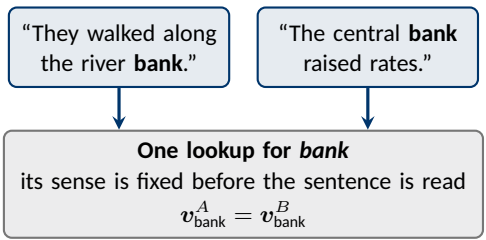


Distribution across judges of the t-statistic from the name-validation regression, by size of the judge's corpus; vertical line at 1.96.

What static vectors still cannot do

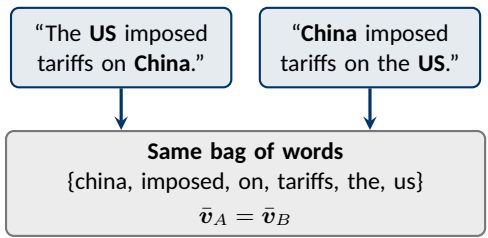
- **What they fixed:** words used alike now sit near each other, so *tariff* and *tax*, *carriage* and *automobile* are no longer orthogonal.
- **What they did not:** a word is still one fixed vector, and a document still an average.

1. One vector per word (polysemy)



Two meanings:
a riverside vs. a monetary authority.

2. No order (same bag of words)



Opposite shocks:
a US tariff vs. Chinese retaliation.

→ Recovering meaning requires a representation that reads each word **in its sentence**: which words appear near which, and in what order.

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Contextualized Sequence Embeddings

The Polysemy Problem

Static vectors fail when one word has multiple meanings.

- **Context A**: “The **bank** of the river.”
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Self-Attention Mechanism (first pass)

For a word i, the contextualized vector $\mathbf{h}_i$ is a weighted sum over all words j in the sequence:

$$\mathbf{h}_i = \sum_j \alpha_{ij}\, \mathbf{z}_j$$

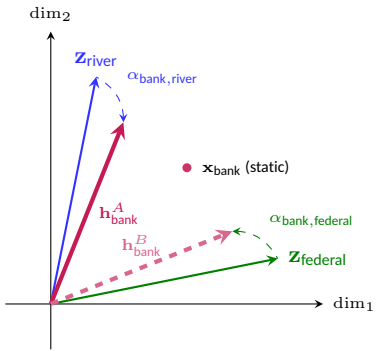
where $\mathbf{z}_j$ is the representation of word j entering the layer (defined two slides on) and α_{ij} is the weight word i places on word j.

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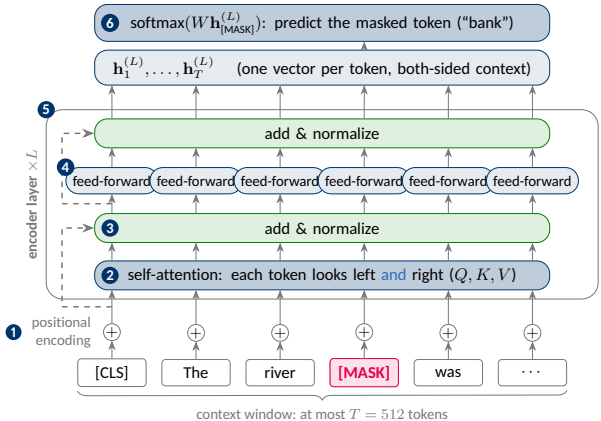
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The transformer is the function f

Vaswani et al., 2017, “Attention Is All You Need”

A transformer is the function f in $\mathbf{h} = f(\text{word}, \text{context})$, built by stacking:

1. **Positions**: take the order of each word into account.
2. **Self-attention**: take the surrounding words into account (queries, keys, values).
3. **Add & Norm**: residual connection and layer normalization.
4. **Feed-forward**: transform each word on its own (the nonlinearity).
5. **Stack**: repeat steps 2–4 L times ($L = 12$ in BERT-base).
6. **Predict text**: guess the masked word; this estimates every weight.



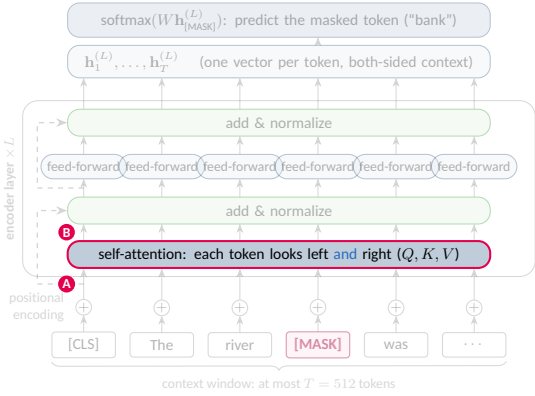
Self-attention: queries, keys, values

Three learned linear maps of the layer's inputs give each word a **query** $\mathbf{q}_i = \mathbf{W}_Q \mathbf{z}_i$, a **key** $\mathbf{k}_j = \mathbf{W}_K \mathbf{z}_j$ and a **value** $\mathbf{v}_j = \mathbf{W}_V \mathbf{z}_j$; the average now runs over the values $\mathbf{v}_j$, not $\mathbf{z}_j$ itself ($\sqrt{d}$ keeps the softmax from getting too sharp as d grows):

$$\alpha_{ij} = \frac{\exp\left(\mathbf{q}_i^\top \mathbf{k}_j / \sqrt{d}\right)}{\sum_l \exp\left(\mathbf{q}_i^\top \mathbf{k}_l / \sqrt{d}\right)}$$

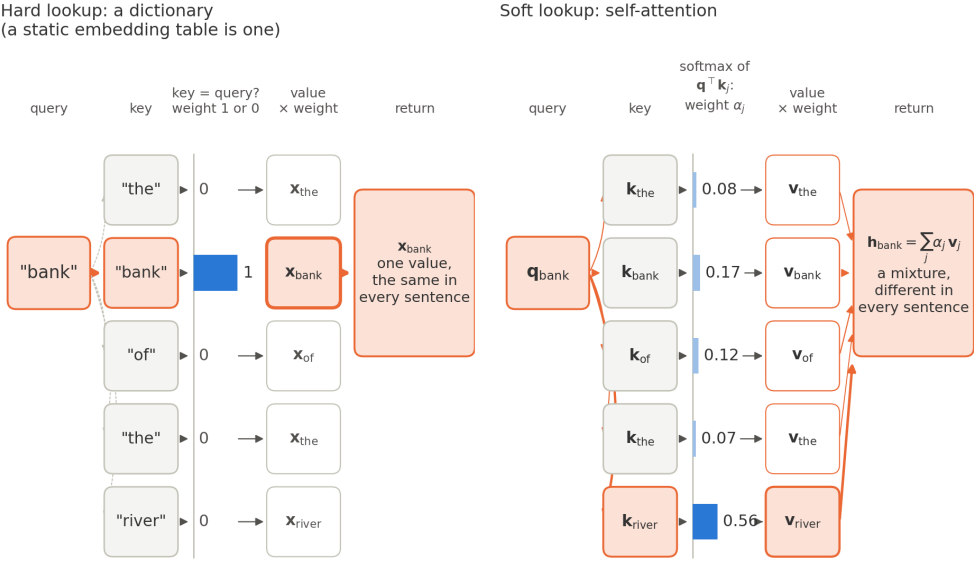
$$\underbrace{\mathbf{h}_i}_{\text{B}} = \sum_j \alpha_{ij}\, \mathbf{W}_V \underbrace{\mathbf{z}_j}_{\text{A}}$$

- **A input** $\mathbf{z}_1, \ldots, \mathbf{z}_T$: every token's vector in the window (embedding + position); unlike the FFN, the layer reads all of them at once.
- **B output** $\mathbf{h}_i$: a weighted average of the values; one new vector per token, passed up to add & normalize.



Attention as a soft dictionary lookup

(Murphy, 2022; Meng, 2024)



Attention alone is a bag of words: positional encoding

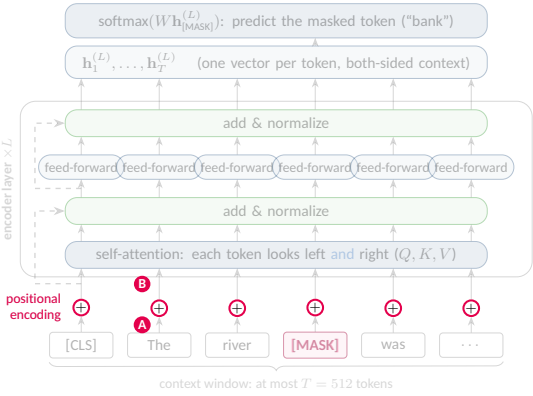
Self-attention on its own **does not know where words are**: $\mathbf{h}_i = \sum_j \alpha_{ij}\mathbf{v}_j$ averages over a *set* of vectors, with weights that depend only on the vectors themselves. Shuffle the words and the outputs shuffle with them.

The fix: put position into the input. Word i enters as

$$\underbrace{\mathbf{z}_i}_{\text{B}} = \underbrace{\mathbf{x}_i}_{\text{A}} + \mathbf{p}_i$$

- **A word embedding** $\mathbf{x}_i$: a static vector as before, now learned jointly with the rest.
- **B output** $\mathbf{z}_i$: the word vector plus $\mathbf{p}_i$, which encodes "position i": the one place order enters.

Both repairs are now in place: word vectors for *different words*, *same meaning*; positions for *same words*, *different order*.



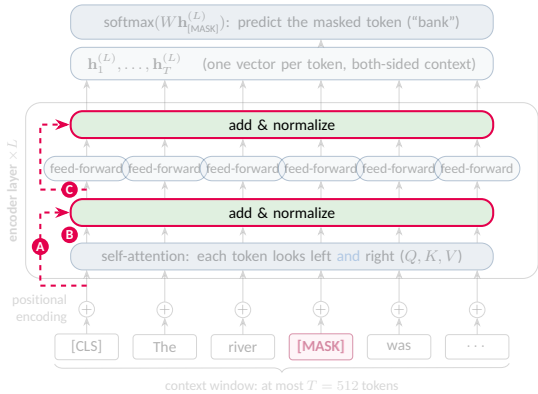
Add & normalize: what makes a deep stack estimable

Each sublayer F (attention, then FFN) is wrapped the same way; for token i:

$$\underbrace{\mathbf{h}_i}_{\text{C}} = \mathrm{LN}\Big(\underbrace{\mathbf{x}_i}_{\text{A}} + \underbrace{F(\mathbf{x})_i}_{\text{B}}\Big)$$

- **A input** $\mathbf{x}_i$: enters F and skips around it on the dashed path.
- **B sublayer output** $F(\mathbf{x})_i$: what attention (or the FFN) computed.
- **C output** $\mathbf{h}_i$: the sum, rescaled, handed to the next block.

Add: F learns a correction to $\mathbf{x}_i$, not a replacement; the identity path carries gradients to early layers (He et al., 2016).
Normalize: $\mathrm{LN}(\mathbf{u}) = \gamma \odot \frac{\mathbf{u}-\mu}{\sigma} + \beta$, with μ, σ over the d entries of $\mathbf{u}$: every token sits on the same scale at every layer.



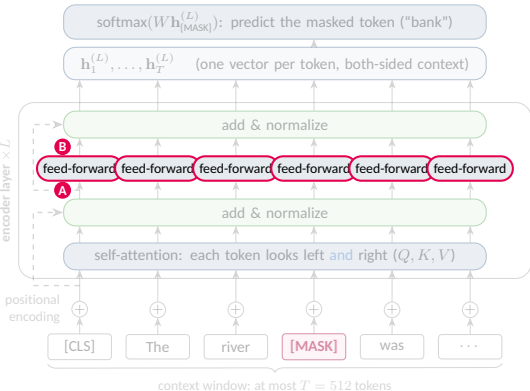
Feed-forward Layer

The sublayer F after attention is a two-layer network applied to one token at a time:

$$\underbrace{F(\mathbf{x})_i}_{\text{B}} = W_2\,\sigma\big(W_1\,\underbrace{\mathbf{x}_i}_{\text{A}} + \mathbf{b}_1\big) + \mathbf{b}_2$$

- **A input** $\mathbf{x}_i$: this token's vector after attention and add & normalize.
- **B output** $F(\mathbf{x})_i$: a new vector for the same token, passed up.

Why it is there: attention only averages; the FFN adds the nonlinearity. Attention picks which tokens to combine, the FFN what to make of them. Most parameters, and most stored facts, live here (Geva et al., 2021).

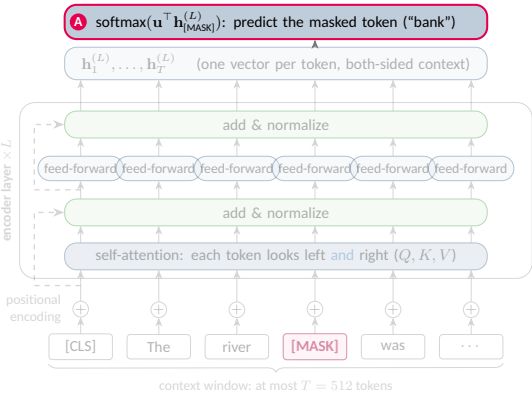


How it is estimated: predict a word, then a label

Pretraining: predict text. None of these parameters is set by hand. As with Word2Vec, θ is fit by predicting words, but now from the context-dependent vector rather than a single lookup (output vectors $\mathbf{u}_w$, as in skip-gram). **GPT** (Radford et al., 2018) predicts the next word from the last position:

$$P(w_{t+1} \mid w_1, \ldots, w_t) \propto \exp\left(\mathbf{u}_{w_{t+1}}^\top \mathbf{h}_t^{(L)}\right)$$

BERT (Devlin et al., 2019) instead hides a fraction of the words and predicts each one from both sides, A the masked position. Either way, fit by $\max_\theta \sum_t \log P$ on unlabeled text: the corpus is the data, no labels needed.



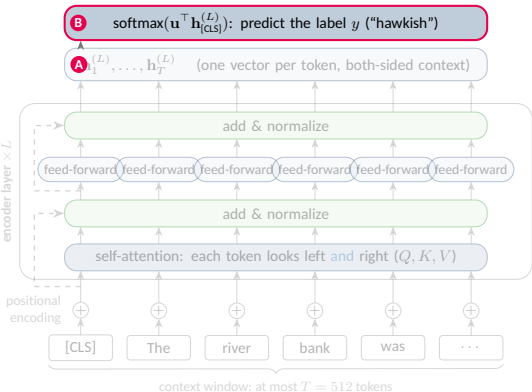
How it is estimated: predict a word, then a label

Supervised: predict a label. Same network, new target. Nothing is masked; the top layer predicts a label y (hawkish/dovish, topic) from one pooled vector per document, Ⓐ the [CLS] position:

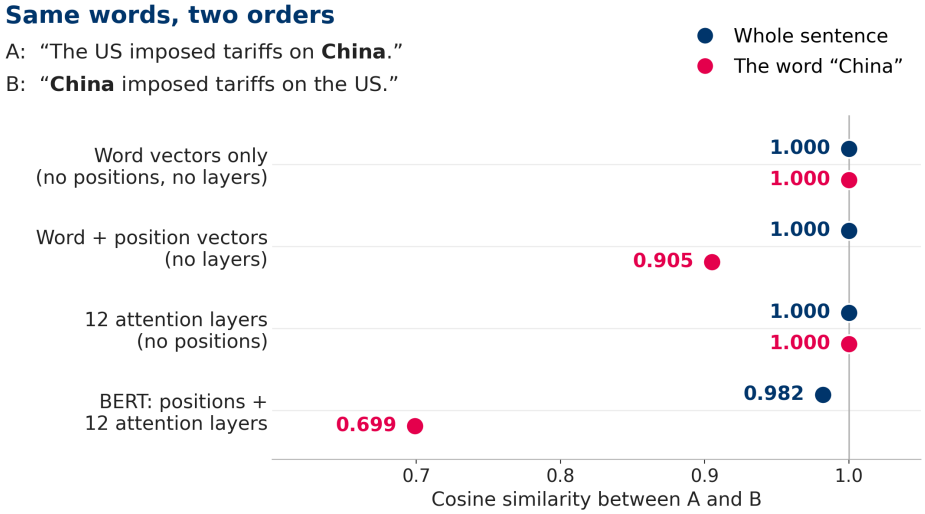
$$P(y \mid \text{doc}) = \text{softmax}\big(\mathbf{u}^\top \mathbf{h}^{(L)}_{\text{[CLS]}}\big)_y$$

Fit by $\max \sum_i \log P(y_i \mid \text{doc}_i)$ on labeled documents, in one of two ways:

- Ⓑ **Linear probe**: freeze θ, fit only $\mathbf{u}$. A logit on pretrained features; a few hundred labels suffice.
- **Fine-tuning**: update θ too, from its pretrained value: pretraining is the initialization, the labels are the data.



Which ingredient does what: word order



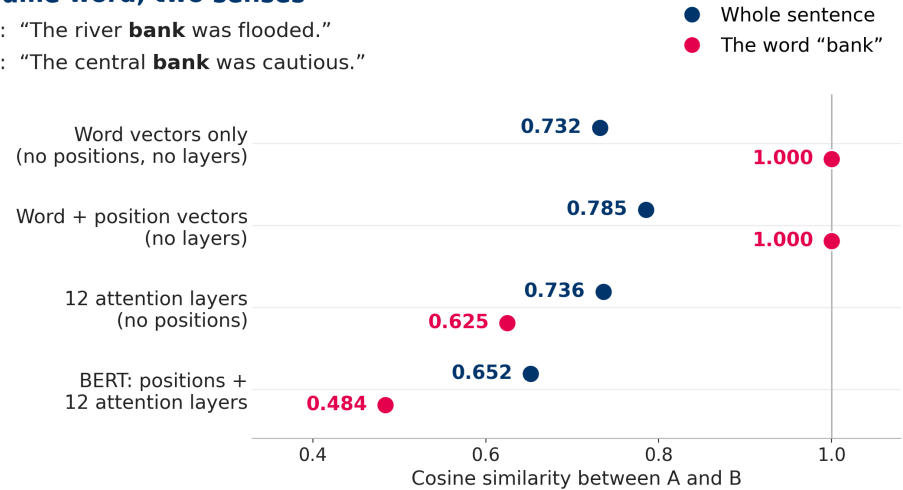
- Four versions of BERT-base, switching the pieces off one at a time. **Order** is detected only in the last row: positions tag each token with its slot, but only attention reads the tags, and attention without positions is permutation-blind.

Which ingredient does what: word sense

Same word, two senses

A: “The river **bank** was flooded.”

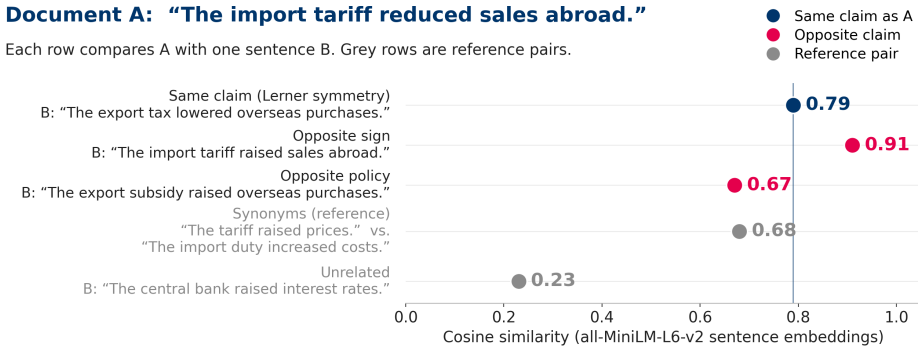
B: “The central **bank** was cautious.”



- **Sense** is resolved as soon as attention is present, with or without positions, because attention can read which other words are in the sentence; positions only sharpen the split

Then, contextual embeddings are perfect?

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- `all-MiniLM-L6-v2`, a sentence-embedding model (a contextual transformer fine-tuned so that sentences with similar meanings land close; Reimers & Gurevych, 2019), lifts the Lerner pair from 0 under TF–IDF to **0.79**.

- But Document A’s own reversal, *the import tariff **raised** sales abroad*, scores **0.91**: the sentence that contradicts A sits closer than the one that agrees. The model has captured **topic**; which way anything moves barely registers.

Representations of language are not representations of economics

- Lerner symmetry is a fact of a different kind: a **theorem about general equilibrium**, not a regularity in how the phrases are used. Nothing in the text marks the Lerner pair as the one that agrees.
- Contextual models do not change this (they resolve polysemy, a fact about usage; the sentence model above is itself contextual), nor will more text of the same kind. What would is a training signal that encodes the economics: pairs labelled *these two make the same claim*. **That is not more data; it is a criterion, and it has to come from the researcher.**
- **Asking an LLM directly** is a different instrument: closer to *recalling* a textbook result than to *measuring* similarity. It need not extend to an equivalence specific to your question, and its criteria are not stated, so you cannot audit why it judged two documents alike.

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The Decision Space

The Constraints

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- **Inference:**
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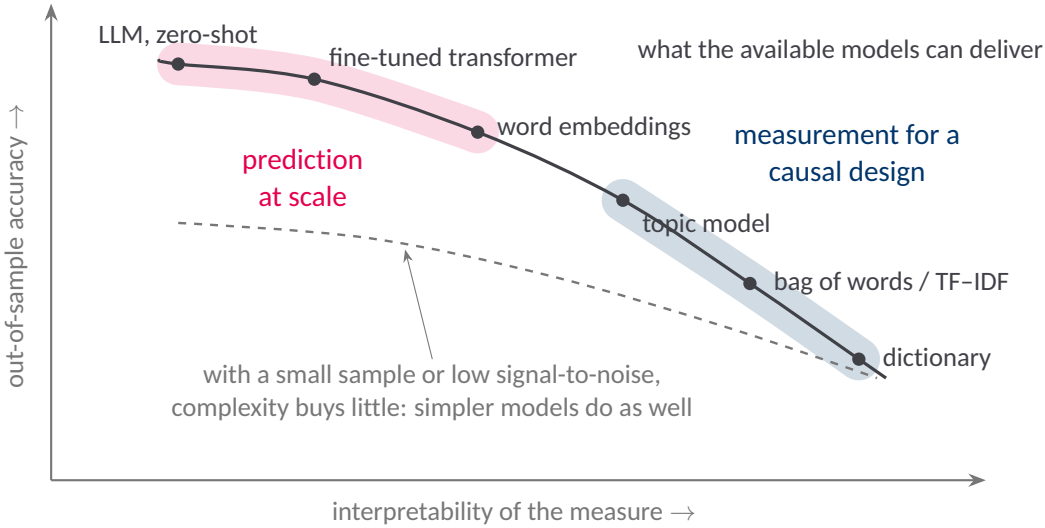
Downstream (Tasks)

- **Measurement:**
  Quantifying concepts.
- **Inference:**
  Causal estimation / Prediction.

The Constraints

- **Research Goal:** Do you need interpretability or accuracy?
- **Data:** Sample size and signal-to-noise ratio.
- **Resources:** Budget (Compute/APIs) and Time.

The choice of representation as a trade-off



Measuring High-Dimensional Concepts with Unstructured Data

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- **Document Similarity**: uses the distance between two documents to proxy distance in an economically relevant space.
  - Patent filings $\rightarrow$ novelty and influence of technologies (Kelly et al., 2021);
  - College syllabi and academic journal articles $\rightarrow$ the gap between course content and the newest research (Biasi & Ma, 2022);
  - 10-K product descriptions $\rightarrow$ how closely two firms compete, and new industry boundaries (Hoberg & Phillips, 2016).

Methodology: Inference with Complex Data

- **Causal Inference**: control for unobserved confounding in unstructured text data.
  - Integrate *BERT* and *Doc2Vec* embeddings of job postings as high-dimensional nuisance parameters in double machine learning (Ahrens et al., 2025).

- **Demand Estimation**: structural model for creatively differentiated products.
  - Random coefficient nested logit with embeddings of font images (Han & Lee, 2026).
  - $\rightarrow$ Captures strictly *local competition*; diversion ratios decline sharply with visual distance, whereas non-embedding models yield flat, non-local substitution.

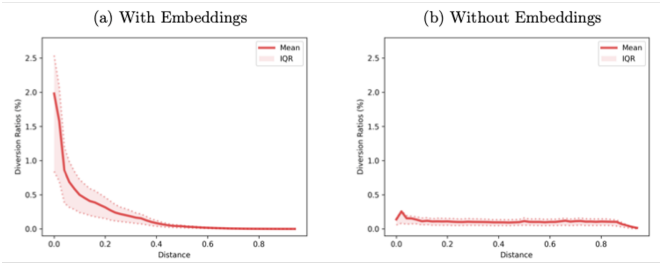


Figure: Price Diversion Ratio and Original Embedding Distance

Upstream: Choose An Embeddings

- Training Strategy: Scratch vs. Pre-trained
  - **Pre-trained Models:** Leverage "world knowledge" encoded in parameters from massive datasets. Modern LLMs/Vision models have already learned the *universal structures* of language or images, requiring only **Fine-Tuning** for PE tasks.
  - **From scratch:** cheaper to understand, and the representation is a function of *your* corpus alone, which is exactly what Ash, Chen & Ornaghi needed to measure each judge's own associations.

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- Domain Adaptation
  - **Context:** Does the model understand politics? (e.g., *PoliBERT* vs. standard *BERT*).
  - **Architecture:** Transformers excel at *long-range dependencies*, while RNNs/LSTMs may suffice for simple sequential scaling.

Thank you!

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